

Lambda calculus

Exercise sheet 5

Exercise 5.1. Let $ID = \lambda x.x$ and $\Omega = (\lambda x.xx)(\lambda x.xx)$. Reduce the following terms to normal form:

- (a) $(\lambda xy.yx) z ID$
- (b) $(\lambda x.\lambda y.xy) y$
- (c) $TRUE ID \Omega$
- (d) $\bar{1} \bar{2} \bar{3}$

Exercise 5.2. Define as lambda terms:

- (a) disjunction;
- (b) implication;
- (c) exclusive disjunction.

Exercise 5.3. Define the following functions for Church numerals:

- (a) zero test: $\bar{0} \mapsto TRUE$, $\bar{0} \neq \bar{n} \mapsto FALSE$;
- (b) parity test: $\overline{2k} \mapsto TRUE$, $\overline{2k+1} \mapsto FALSE$;
- (c) exponentiation: $\bar{n}, \bar{m} \mapsto \bar{n}^m$.

Exercise 5.4. Give explicit definitions, i.e. without relying on previously defined terms, for:

- (a) addition;
- (b) multiplication;
- (c) * exponentiation (assuming η -equivalence, i.e. the rule $\lambda x.Mx \equiv_{\eta} M$, provided x does not occur free in M).